

The Coin Flip Judge? Reliability and Bias in LLM-as-a-Judge Evaluation

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Abstract

LLM-as-a-Judge is widely used to rank model outputs and populate leaderboards, yet its run-to-run reliability is under-characterized. We study repeated identical evaluations across **six judges from three providers spanning three model generations (2024-2026)**, with 50 pairwise and 50 pointwise trials per question on 29 tasks. Three findings stand out. First, pairwise verdicts are unstable, with an 11-14% mean flip rate on identical re-runs, and this has *not* improved across generations: the newest, cheapest deployable judges flip just as often as 2024 models. Second, a dedicated swap-order experiment (8,700 judgments) shows the fixed-order asymmetry is *order bias*, not self-preference, and that order bias has not been fixed either: the 2026 judges reverse its *direction*, favouring the second slot where earlier judges favoured the first, so a first-position check records them as unbiased while they still decide 21-41% of matched comparisons on slot rather than content. Third, forced-choice verdicts and scalar scores come apart, and ~ 11 repeated trials are needed to stabilize a verdict. We argue that multi-trial aggregation, symmetric order randomization, and uncertainty reporting should be standard practice.

1 Introduction

LLM-as-a-Judge is now central to modern evaluation: LLMs rank model outputs, approximate human preferences in benchmark construction, and serve as reward proxies in RLHF (Zheng et al., 2023; Dubois et al., 2024). This makes automated evaluation cheap and scalable, but raises a basic measurement question: *if we ask the same judge the same question multiple times, do we get the same answer?* This is distinct from the better-studied question of whether judges are *biased* (Wang et al., 2023a; Stureborg et al., 2024); we study *repeated-trial reliability*, the extent to which a fixed judge, given identical inputs under nominally identical

conditions, returns the same verdict. It matters for grounding evaluation: if verdicts fluctuate across identical re-runs, single-trial leaderboards and paper tables are unstable even when the model outputs are held fixed.

We run 50 pairwise and 50 pointwise trials per question on 29 tasks across 10 categories, across six judges from three providers spanning three model generations (2024-2026), plus a swap-order experiment and temperature and prompt ablations. Five messages emerge. First, pairwise judging is often noisier than it appears (mean flip rate 11-14%). Second, pairwise verdicts and pointwise scores *come apart*: judges pick winners even when their own scalar scores are near-identical. Third, a swap-order experiment shows the fixed-order asymmetry is *position bias*, not self-preference. Fourth, repeated voting stabilizes verdicts predictably but slowly (~ 11 trials). Fifth, and most striking, the newest, cheapest deployable judges (2026) flip just as often as the 2024 models, and order dependence has not gone away either: it has changed *direction*, with the newest judges locking onto the second slot where earlier ones locked onto the first.

Contributions. (i) A 50-trial-deep reliability characterization of LLM judges on fixed, competitively-matched items, quantifying run-to-run flip rates, an intraclass-correlation view of pointwise scoring, and a principled trial-count recommendation. (ii) A swap-order experiment that cleanly separates position bias from self-preference. (iii) A *cross-generation* panel of efficient judges (2024-2026, three providers), showing that neither run-to-run instability nor order dependence has improved, and that a first-slot-only bias metric hides the second effect. An earlier preprint of ours (Yagubyan, 2026) reports the 50-trial protocol of (i) on two same-provider judges under a fixed presentation order; (ii) and (iii) are new here.

2 Related Work

LLM-as-a-Judge and its biases. Zheng et al. (2023) introduced MT-Bench and the LLM-as-a-Judge paradigm, now standard for open-ended evaluation and preference data (Dubois et al., 2024). Wang et al. (2023a) catalogued position, verbosity, and self-enhancement biases and proposed swap-augmented evaluation, and Shi et al. (2024) study position bias at scale, but both evaluate a *single trial* per condition, which cannot separate systematic bias from run-to-run noise. Panickssery et al. (2024) show judges can favor their own generations; our swap-order design isolates this self-preference confound from pure position effects and finds, on our data, that position, not self-preference, drives the fixed-order asymmetry.

Relation to our earlier preprint. An earlier preprint of ours (Yagubyan, 2026) establishes the 50-trial protocol on the same 29 items for two same-provider judges under a fixed presentation order, reporting a 13.6% mean flip rate and a 72% first-slot preference for GPT-4o-mini. That preprint explicitly flags the self-preference confound inherent in its fixed-order design and identifies a controlled swap study with out-of-family responses as the necessary next step. This paper carries out that study (§4.4), which finds the asymmetry is order bias with self-preference undetectable, and extends the panel to six judges across three providers and three generations. The flip-rate characterization in contribution (i) is therefore already public; contributions (ii) and (iii) are new here.

Reliability and determinism. Closest to us, Haldar and Hockenmaier (2025) establish low intrarater reliability for LLM judges across repeated runs, spanning both direct-assessment scoring and pairwise preference labels, and show that aggregating multiple runs improves agreement with human judgements; Atil et al. (2024) show that LLM outputs are non-deterministic even at temperature zero. Our delta is threefold. We measure at 50-trial depth on fixed, competitively-matched items, which is what makes per-item flip-rate distributions and a reliability curve estimable; we add a controlled swap experiment that separates order bias from self-preference, which their design does not test; and we run a cross-generation panel that asks whether newer models resolve either problem. Our framing borrows from psychometrics, where repeated-measure reliability is quantified by the in-

traclass correlation (Cicchetti, 1994; Koo and Li, 2016); majority-vote aggregation over trials is the same mechanism as self-consistency (Wang et al., 2023b), applied here to the judge rather than the solver.

3 Method

Problem setup. Fix a judge J , a question q , and a pair of candidate responses (a, b) . A single pairwise evaluation draws a verdict $v \sim J(q, a, b) \in \{A, B, \text{tie}\}$; because decoding is stochastic, v is a random variable even with all inputs held fixed. We are interested in the *dispersion* of this distribution, not merely its mode. Running N independent trials yields counts $(n_A, n_B, n_{\text{tie}})$, and we summarize instability by the flip rate $1 - \max(n_A, n_B, n_{\text{tie}})/N$: the fraction of trials that disagree with the majority verdict. A flip rate of 0 means the judge is perfectly reproducible on that item; a flip rate approaching 0.5 (for two-way outcomes) means its verdict is close to a coin toss. All downstream metrics below are functions of these per-item trial distributions.

Data and judges. We use 29 competitively-matched question-response pairs spanning 10 task categories. Responses A and B are generations from GPT-4o-mini and GPT-4o respectively (deliberately close in quality, a stress test). Our primary judges are gpt-4o-mini and gpt-4.1-mini; a cross-provider replication adds gemini-2.5-flash, and to test the newest generation we additionally run three 2026 efficient judges from two providers (gpt-5.4-nano, gpt-5.4-mini, claude-haiku-4.5), six judges across three providers in total. The main experiment uses default temperature ($t=1.0$); an ablation uses $t=0$.

Protocol. (E1) For each (judge, question) we run 50 pairwise comparisons and 2×50 pointwise (1-10) scorings. *Pairwise order is held fixed* (Response A first every trial); we flag this prominently, as it isolates pure sampling variability but confounds position with content and self-preference in the raw A-rate. (E2) A temperature ablation repeats the pairwise protocol at $t=0$ (10 trials). (E3) A prompt-sensitivity ablation uses a second, semantically equivalent template. (E4) A *swap-order* experiment re-judges every question in *both* presentation orders (30 trials/order/judge; 1,740 judgments per judge, 8,700 in total across the five-judge panel), which cleanly separates position, content,

and self-preference. **(E5)** A third-provider judge runs the full E1 protocol.

Metrics. **Flip rate** is $1 - \max(n_A, n_B, n_{\text{tie}})/N$. **Pairwise-pointwise gap** (PPG) is $|\bar{s}_A - \bar{s}_B|$. **First-slot preference** (FSP) is the fraction of judgments won by the first-shown response. Because a judge may lock onto *either* slot, we also report a symmetric **order-driven rate**: the fraction of paired trials on which the judge picks the same slot in both presentation orders, first or second. A first-slot-only statistic is blind to a second-slot preference and can report a reversed bias as a fix. We report **ICC(2,1)** for repeated pointwise scores (convention: <0.60 poor-to-moderate), cross-judge **Cohen’s κ** , and a **reliability curve** $P(\text{majority}(K) = \text{majority}(50))$ by Monte-Carlo subsampling. Nonparametric tests are used throughout; CIs are 10,000-sample bootstraps over items. Verdicts are three-way, so the reference for an uninformative judge is not 50%: a uniform $\{A, B, \text{tie}\}$ judge at $N=50$ has an expected flip rate of 59.7% (limiting value 66.7%), and we use that as the coin-flip reference throughout.

4 Results

4.1 Pairwise judging is unstable

All judges flip 11-14% of the time on average (Table 1), i.e. roughly one in seven identical reruns would change. The distribution is bimodal: many questions are perfectly stable (up to 14/29 at $\text{FR}=0$), while others approach the coin-flip reference: the worst single item reaches 65% (Claude Haiku 4.5). Because verdicts are three-way, per-judge tie rates (2.6-14.2%, Table 1) let an item exceed the two-outcome ceiling of 50%; the relevant reference is 59.7% (§3).

This instability persists across two providers and three model generations (Figure 2, left), but the panel is small and we state the claim with its uncertainty. At $n=29$ items the bootstrap 95% CI on each judge’s mean flip rate spans roughly six points either side (Table 1), so the 2.9-point spread across the five judges lies inside every interval. Because all judges saw the same items, the paired per-item comparison is sharper: GPT-4o-mini (2024) minus each 2026 judge gives $+0.0$, $+2.0$ and $+2.3$ points, with 95% CIs of $[-7.6, +7.5]$, $[-6.4, +10.4]$ and $[-7.0, +11.2]$. This is a *bounded-power null*: we can exclude an improvement larger than roughly 8-11 points, not demonstrate equivalence. What the

Judge	Mean FR [95% CI]	Max FR	Tie rate	ICC	PPG	Order- driven
4o-mini	13.3 [7.7, 19.6]	46	2.6	0.58	0.25	46.0
4.1-mini	13.9 [7.9, 20.3]	56	10.5	0.77	0.36	23.1
5.4-nano	13.3 [7.5, 19.6]	60	10.7	0.70	0.23	41.3
5.4-mini	11.3 [6.3, 16.9]	55	11.6	0.81	0.33	33.9
haiku-4.5	11.0 [5.4, 17.5]	65	7.5	0.64	0.19	21.1

Table 1: Aggregate consistency for all five judges ($t=1.0$); judge names abbreviated as in Table 4. FR = flip rate (%) with a 10,000-sample bootstrap CI over the 29 items; Tie rate = share of three-way tie verdicts; PPG = pairwise-pointwise gap; Order-driven = symmetric rate at which the judge picks the same slot in both presentation orders (§4.4). Every flip-rate interval overlaps every other: neither instability nor order dependence is resolved in the newest models.

data do rule out is the generational fix one might expect.

4.2 The pairwise-pointwise gap

Our central finding is a disconnect between forced-choice verdicts and scalar scores. Mean pointwise gaps are tiny (0.25 and 0.36 points on a 10-point scale) and not significant in aggregate (Wilcoxon $p=0.88$, $p=0.13$), though at $n=29$ this is a bounded-power null, not demonstrated equivalence. Yet in pairwise mode the same judges still confidently pick winners. A variance decomposition on the pointwise scores shows within-pair (stochastic) noise is strongly judge-dependent, ranging from 9.5% of pointwise variance (GPT-4.1-mini) to 42% (GPT-4o-mini), with the three 2026 judges in between (17-34%); this tracks the judges’ ICC values (0.58-0.81 in Table 1). A single “8 vs 9” pointwise score therefore cannot be read as a reliable quality difference for the noisier judges, and this holds across all five.

We note the natural alternative reading. PPG is computed from pointwise scores alone and contains no pairwise term, so a near-zero gap beside a confident verdict need not indicate a pathological judge: a 1-10 integer scale on which judges compress almost all mass into 7-9 may simply be too blunt to resolve a difference that a forced choice can. Our data do not separate these two accounts, and the practical consequence is the same either way. Whichever mechanism holds, a sub-point pointwise gap does not license the ranking that the same judge’s pairwise verdict asserts, so the two modes should not be treated as interchangeable measurements of the same quantity.



Figure 1: Per-question pairwise flip rate (%) for all five judges, one row per judge, columns sorted by mean instability across judges. The pattern is bimodal: a large stable head (green, many items near 0%) and a hard tail (orange/red) approaching the “coin-flip” band. Instability concentrates on the same discriminative subset of items across judges.

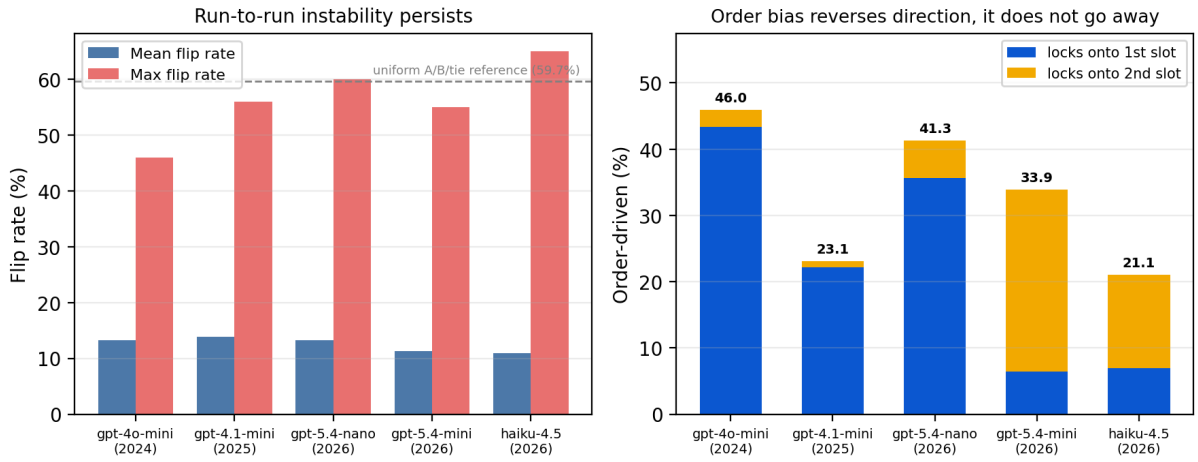


Figure 2: Five efficient judges across two providers and three generations (2024-2026); both panels computed from the released judgments. **Left:** run-to-run flip rate (mean and worst-case) is essentially unchanged across generations; the dashed line is the 59.7% expected of a judge drawing uniformly over $\{A, B, \text{tie}\}$ at $N=50$. **Right:** symmetric order-driven rate from the swap-order experiment, split by which slot the judge locks onto. The 2026 judges (GPT-5.4-mini, Haiku 4.5) have shed *first*-slot bias but acquired second-slot bias; totals stay at 21-41%, and the 2025 GPT-4.1-mini is already as low as the best 2026 model.

4.3 Which tasks are unstable

Instability is not uniform across task types, but at 2-4 items per category this analysis is *exploratory* and we report it as such. A Kruskal-Wallis test over the ten categories is not significant ($H=15.4$, $p=0.082$, $n=29$), and we deliberately report no effect size: for $k=10$ groups and $n=29$ the null expectation of the usual rank η^2 is $(k-1)/(n-1)=0.32$, so any value near that number is uninformative rather than large. What survives is the coarse ordering in Table 2: coding is the least reproducible category by a clear margin (28.3% mean flip rate), math is next (18.8%), and the remaining eight categories cluster between 4.7% and 15.2% with overlapping spreads. No pairwise category contrast survives multiple-comparison correction,

and none of the paper’s claims rests on one. Stratifying by difficulty, items that flip at all ($FR \geq 10\%$) average a 23.6% flip rate, versus 2.9% for the stable majority, so the instability is concentrated in a hard subset rather than spread evenly, which is exactly where a benchmark’s discriminative items tend to live. This matters for practice: the questions most likely to separate two competing systems are also the questions a judge is least able to score reproducibly.

A concrete case. The single most unstable item in our set is a reasoning question on which GPT-4.1-mini flips 56% of the time, close to the 59.7% expected of a judge drawing uniformly over $\{A, B, \text{tie}\}$. The two candidate answers reach the same conclusion by slightly different routes, and

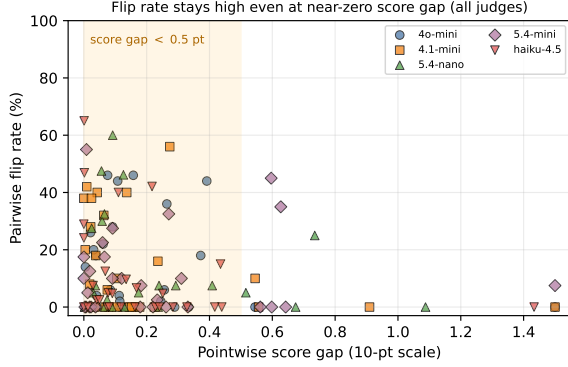


Figure 3: Pairwise flip rate versus pointwise score gap per question, all five judges. High flip rates persist even where the two responses receive nearly identical scalar scores (shaded band, gap < 0.5 pt): the pairwise-pointwise gap. Two items with gap > 1.5 are clipped.

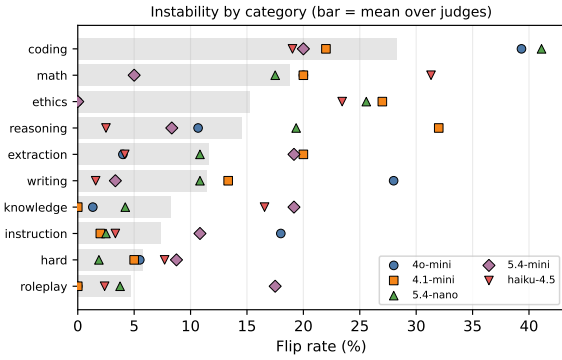


Figure 4: Flip rate by task category for all five judges (points), with the mean over judges as the bar. Open-ended, reasoning-heavy categories are least reproducible; closed categories (knowledge, roleplay) most.

the judge has no stable basis for preferring one, so its verdict is effectively resampled on each call. Reporting a single verdict for such an item is not measuring quality; it is reporting one draw from a near-uniform distribution. A multi-trial protocol would at minimum surface that the item is undecidable for this judge.

4.4 Order bias, not self-preference

Under fixed order, GPT-4o-mini prefers the first slot on 72% of questions ($p=0.024$); this confounds position, content, and family self-preference (Response A is always GPT-4o-mini’s own output). The swap-order experiment (E4) resolves it (Table 3). With content balanced across slots, GPT-4o-mini selects whatever occupies the *first slot* on 69.4% of 1,740 judgments ($p \approx 10^{-60}$); on 43% of paired trials it picks the first slot in *both* orders (first-slot-driven judging).

Category	4o-m	4.1-m	5.4-n	5.4-m	haiku
coding	39.3	22.0	41.1	20.0	19.0
math	20.0	20.0	17.5	5.0	31.3
ethics	0.0	27.0	25.6	0.0	23.4
reasoning	10.7	32.0	19.4	8.3	2.5
extraction	4.0	20.0	10.8	19.2	4.2
writing	28.0	13.3	10.8	3.3	1.6
knowledge	1.3	0.0	4.2	19.2	16.6
instruction	18.0	2.0	2.5	10.8	3.3
hard	5.5	5.0	1.9	8.8	7.7
roleplay	0.0	0.0	3.7	17.5	2.4

Table 2: Mean flip rate (%) by task category for all five judges, all ten categories, rows ordered by the mean across judges. Coding (28.3) and math (18.8) lead; the remaining categories fall between 4.7 and 15.2 with overlapping spreads. Small per-category n (2-4) means these are exploratory and no category contrast is significant.

Content consistency under swap is at coin-flip level (50.1%): reversing order erases the verdict about half the time. Crucially, among content-consistent pairs the judge prefers its own output only 52.8% of the time; *self-preference is undetectable* ($p=0.27$). The fixed-order asymmetry is position bias. GPT-4.1-mini shows the same effects at roughly half the magnitude (54.6% first-slot).

A first-slot-only metric is not enough. A judge can lock onto *either* slot, and the *position-driven* column counts only first-slot lock-in, so it cannot see a second-slot preference. Table 3 therefore reports both directions and their sum. The distinction is not hypothetical: GPT-5.4-mini is first-slot-driven on just 6.5% of paired trials but *second*-slot-driven on 27.4%, for a symmetric order-driven rate of 33.9%. Its first-slot preference of 33.5% is a 16.5-point deviation from the unbiased 50%, the second largest in the panel. Reporting only the first-slot figure would record this judge as nearly unbiased.

Order bias across generations: direction, not magnitude. Running the same swap-order protocol on the 2026 judges shows that what changed across generations is mainly the *direction* of the order effect, not its size. The symmetric order-driven rate falls from 46.0% (GPT-4o-mini, 2024) to 41.3%, 33.9% and 21.1% for the three 2026 judges (Table 3, Figure 2, right), but the 2025 GPT-4.1-mini already sits at 23.1%, as low as the best 2026 model, so the sequence is not monotone in release date. Meanwhile the *sign* flips: GPT-4o-mini, GPT-4.1-mini and GPT-5.4-nano over-select the

Judge	First-slot	1st-slot driven	2nd-slot driven	Order-driven	Content-cons.
4o-mini	69.4	43.3	2.6	46.0	50.1
4.1-mini	54.6	22.2	0.9	23.1	58.7
5.4-nano	59.7	35.7	5.6	41.3	42.0
5.4-mini	33.5	6.5	27.4	33.9	50.2
haiku-4.5	39.8	6.9	14.1	21.1	66.2

Table 3: Swap-order decomposition across all five judges, in % of the 1,740 judgments (870 paired trials) per judge. *First-slot*: share of judgments won by the first-shown response (50% = unbiased). *1st-/2nd-slot driven*: picks that same slot in *both* orders. *Order-driven*: their sum, the symmetric rate at which slot rather than content decides the verdict. *Content-cons.*: verdict survives an order swap. Self-preference, defined only for GPT-4o-mini and GPT-4.1-mini (which authored Responses A and B), is undetectable at 52.8% and 53.0% ($p=0.27, 0.18$); the fixed-order asymmetry is order bias, not self-enhancement. The newest judges reverse the direction of the effect rather than removing it.

first slot (FSP 54.6-69.4%), whereas GPT-5.4-mini and Claude Haiku 4.5 over-select the second (FSP 33.5% and 39.8%). On a first-slot-only metric the latter two look almost unbiased; on a symmetric one they remain order-driven a fifth to a third of the time.

This sharpens rather than softens the paper’s message. Both failure modes survive three model generations: run-to-run instability is flat, and order dependence has been redirected rather than removed. An evaluation protocol that merely checks for first-position bias will certify the newest judges as clean while they are still deciding a third of matched comparisons on slot rather than content.

4.5 A cross-provider replication

Our deep panel rests on two same-provider (OpenAI) judges; the Gemini and Anthropic runs are *replications* that test whether the phenomena are an OpenAI artifact, not a balanced multi-provider design. They are not. Cross-judge agreement is only moderate throughout the panel (Table 4). The two OpenAI judges agree at just 75.9% ($\kappa=0.51$), a Gemini 2.5 Flash judge on the full 50-trial protocol reproduces the core instability while flipping *more* (16.1%), and across all five efficient judges pairwise κ ranges from 0.08 to 0.63. Gemini’s own agreement with the panel spans $\kappa=0.23$ -0.63, as wide as the panel’s internal spread, so no judge in this set is a stable reference. The lowest agreement is between two *OpenAI* models a generation apart (GPT-4o-mini vs. GPT-5.4-mini, $\kappa=0.08$), so

Cohen’s κ	4o-m	4.1-m	5.4-n	5.4-m	haiku
4o-mini	.	.51	.31	.08	.25
4.1-mini	.51	.	.63	.41	.59
5.4-nano	.31	.63	.	.53	.54
5.4-mini	.08	.41	.53	.	.32
haiku-4.5	.25	.59	.54	.32	.

Table 4: Cross-judge agreement (Cohen’s κ) on majority verdicts, all five judges. Agreement is only moderate at best and falls as low as 0.08 (GPT-4o-mini vs. GPT-5.4-mini): the choice of judge changes outcomes as much as, or more than, re-running the same judge.

disagreement is not a cross-provider artifact either. Roughly one in four outcomes depends on which judge is chosen, and the pairwise-pointwise gap replicates across providers (Gemini PPG 0.30).

4.6 What a single pointwise score means

Because pointwise scoring is common in practice, we decompose its variance for all five judges. Pooled, 79.4% of pointwise-score variance is genuine between-response signal and 20.6% is within-response stochastic noise, but the split is strongly judge-dependent: within-response noise runs from 9.5% (GPT-4.1-mini) through 17-19% (GPT-5.4-nano, GPT-5.4-mini) and 34% (Claude Haiku 4.5) to 42% (GPT-4o-mini), mirroring the ICC values in Table 1. The pooled within-response standard deviation is ≈ 0.39 points, implying a 95% margin of roughly ± 0.8 points (up to ± 0.9 for the noisier judge) on any *single* scalar score. A reported “8 vs. 9” difference is therefore within the noise floor for the noisier judge, and small score gaps on leaderboards built from single scores should not be read as real quality differences.

4.7 Temperature and prompt sensitivity

We test two natural mitigations; both only partly help, and greedy decoding in particular is *less* effective for the newest judges. Setting $t=0$ sharply reduces flip rates for the 2024-2025 OpenAI judges (to 2.8% and 7.9%) and drives Claude Haiku 4.5 to full determinism (0%). But it barely moves the 2026 GPT-5.4 judges: GPT-5.4-nano and GPT-5.4-mini still flip 12.9% and 11.0% of the time at $t=0$, essentially unchanged from their $t=1$ rates, with 13-15/29 items still flipping. Greedy decoding is therefore not a reliable fix: the very models that most need it (the newest, cheapest judges) benefit from it least, consistent with reported nondeterminism in production inference (Atil et al., 2024). Prompt wording matters too: on the two primary

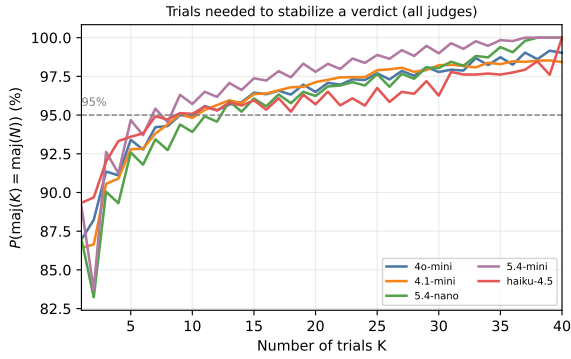


Figure 5: Probability that a K -trial majority matches the full-run majority (in-sample subsampling), for each of the five judges. Even the most stable judge needs several trials to reach 95% agreement; noisier judges need many more.

judges, a second semantically equivalent scoring template changes the *majority* outcome on 25% of tested (judge, question) cells. (We note a caveat: the alternate template introduced a label-mapping ambiguity that caused 47/400 parse failures, so part of this sensitivity may be parsing friction rather than pure wording preference; we therefore report this prompt ablation only for the primary pair.) Neither knob substitutes for multi-trial aggregation.

4.8 How many trials stabilize a verdict?

A single trial matches the full-run majority only about 86% of the time (Figure 5), and the number of trials needed to reach 95% agreement varies across all five judges: from 5 (GPT-5.4-mini) and 9 (GPT-4o-mini) to 11-13 for GPT-4.1-mini and GPT-5.4-nano, with Claude Haiku 4.5 the noisiest (it does not reach 95% within its available trials on the hardest items). For high-flip questions the requirement is far larger (up to 29 trials). Because the full-run majority is itself a noisy estimate on unstable items, these are *lower bounds* on the trials needed to recover a judge’s true preference. The practical implication is uniform across the panel: single-trial judging is insufficient, and roughly 10-15 trials are a reasonable default.

5 Discussion and Recommendations

LLM judge reliability is not a single property; it decomposes into stochastic instability, systematic order bias, protocol sensitivity, and cross-judge disagreement, each of which can undermine a benchmark independently. Our results show these components move separately, and that progress on one can be misread as progress overall: the newest judges

have largely shed *first*-position bias while remaining order-driven a fifth to a third of the time in the other direction, and their run-to-run instability is unchanged. An evaluation can be certified “debiased” by the usual first-position check and still be both order-driven and unreproducible.

A leaderboard noise budget. The practical stakes are easiest to see on a leaderboard. If a single-trial judge flips $\sim 14\%$ of comparisons, then two systems whose true win rate differs by less than roughly twice that margin are statistically indistinguishable under single-trial evaluation, yet such narrow gaps are routinely reported as rankings. Cross-judge disagreement compounds this: with raw agreement of only 76% ($\kappa=0.51$) between two judges of the same provider, roughly one comparison in four would be decided differently by an equally reasonable judge. A ranking that a different judge, or the same judge on a re-run, would reorder is not a measurement of model quality so much as a measurement of the judge.

Implications for groundedness and faithfulness metrics. The instruments this community relies on are largely LLM-judged: faithfulness, groundedness, attribution and hallucination scores are typically a judge’s verdict on whether a claim is supported by a source. Those metrics inherit the numbers reported here directly. A single-trial faithfulness label carries the same 11-14% flip rate we measure, so a reported gain of a few points between two retrieval-augmented systems can be entirely within judge noise; a groundedness *rate* aggregated over items inherits the per-item dispersion rather than averaging it away, because the unstable items are the discriminative ones. Order dependence matters here too: a groundedness prompt that always places the source before the claim, or the claim before the source, is a fixed-order protocol of exactly the kind our swap experiment shows can decide a third of matched comparisons. We recommend that groundedness and faithfulness benchmarks report a flip rate alongside the score, balance presentation order, and treat items above a flip-rate threshold as unlabelled rather than forcing a verdict.

Comparison to human annotators. This is not a counsel of despair: human annotators are noisy too, and the field’s response there, namely multiple annotators, reported agreement, adjudication, is exactly what we recommend for judges. The difference is that LLM trials are far cheaper than

human ones, so there is no excuse for single-trial judging in high-stakes settings.

Recommendations. Three practices follow directly and are cheap to adopt. **(1) Randomize order and report it:** our fixed-order 72% asymmetry is precisely what randomization prevents, and the swap experiment shows slot can dominate content on matched items. Audit order bias symmetrically: a first-position check alone certifies GPT-5.4-mini as unbiased when it is order-driven 33.9% of the time in the opposite direction. **(2) Aggregate multiple trials:** we recommend ≥ 10 -20 trials with majority voting, and explicitly flagging high-flip items rather than reporting a single verdict for them. This is a 10-20 $\times$ inference cost, which is real but bounded, and it need not be paid uniformly: lopsided pairs are stable enough for a single trial, so a practical compromise is to run two trials on everything and escalate to the full K only where the two disagree, which concentrates the cost on the small discriminative subset that actually drives the ranking. **(3) Report scores with uncertainty:** a single pointwise score carries a margin of up to ± 0.9 points for a noisy judge, so sub-point gaps should never drive a conclusion. Adopting these would make LLM-judge evaluations substantially better-grounded at negligible additional cost.

A concrete protocol. Concretely, we recommend the following recipe for any high-stakes LLM-judge comparison. **(1)** Run K trials per item with $K \geq 10$ (our reliability curve suggests $K=11$ for 95% agreement with the 50-trial majority on average, more for hard items). **(2)** Randomize presentation order per trial and, for pairwise judging, balance the two orders. **(3)** Report the majority verdict *with* its flip rate, and treat any item above a pre-registered flip-rate threshold (e.g. 20%) as undecided rather than forcing a winner. **(4)** For pointwise scoring, report a mean with a bootstrap interval and refuse to rank systems whose intervals overlap. **(5)** Pin dated model snapshots and log the resolved version per call, so that provider-side drift does not silently change results. None of these steps requires new tooling, and their total cost is a small multiple of a single-trial run, negligible against the cost of publishing an unreproducible ranking.

6 Conclusion

Repeated identical LLM-as-a-Judge evaluations are less reproducible than their single-trial use implies: they flip $\sim 14\%$ of the time, pick winners their own scores do not support, decide a fifth to a half of matched comparisons on slot rather than content (order bias, not self-preference), and need ~ 11 trials to stabilize, patterns that replicate across three providers. Neither failure mode improves across three model generations; the newest judges only reverse the direction of their order bias, which a first-position check would record as a fix. Multi-trial aggregation, symmetric order randomization, and uncertainty reporting should be standard practice.

Threats to validity. Three points deserve emphasis. Our items are deliberately *competitively matched* (responses from two capable models of similar quality), which is the regime where instability is largest; on lopsided pairs judges are far more stable, so our flip rates are upper bounds for easy comparisons and lower bounds for the hard, discriminative comparisons that decide close leaderboard races. Second, we measure verdict *dispersion*, not *correctness*: a perfectly stable judge could be consistently wrong, and a noisy judge could be right on average; reliability is necessary but not sufficient for a good judge. Third, the cross-generation comparison holds the prompt and protocol fixed across models; a model-specific prompt might reduce a given judge’s instability, but the point stands that the *default* use of these judges is unreliable.

Limitations

Our main study uses two same-provider judges and 29 items; the Gemini and Anthropic runs are replications showing the core phenomena are not OpenAI artifacts, not a balanced multi-provider panel, and a full panel at equal depth remains future work. At $n=29$ the cross-generation comparison is a bounded-power null: the paired per-item intervals exclude improvements beyond roughly 8-11 points but cannot establish equivalence. Category-level analyses (2-4 items each) are underpowered, non-significant, and reported as exploratory; we report no category effect size because its null expectation at $k=10$, $n=29$ is close to any value we could observe. Judge API snapshots can drift; we pin dated model snapshots and record the resolved version per call, so our reported runs are well-specified.

The reliability curve is an in-sample convergence estimate and lower-bounds the trials needed to recover a judge’s true preference.

Reproducibility and Ethics

All code, judge prompts, and the complete per-trial raw judgments are included in the supplementary material. To make our runs well-specified, every experiment pins dated model snapshots and records the resolved model version per API call, so provider-side drift is detectable and the protocol is fully described by the settings reported here (50 pairwise and 50 pointwise trials per item; 30 trials per order in the swap experiment). No human subjects or private data are involved: the 29 items are synthetic question-response pairs generated for this study. Our findings are a caution about *how* LLM judges are used, not an argument against automated evaluation; the recommended mitigations (multi-trial aggregation, order randomization, uncertainty reporting) are inexpensive and improve the integrity of the leaderboards and preference datasets that increasingly shape model development.

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A Judge Prompt

The pairwise judge prompt (used for all judges, held identical across the panel) presents the question and the two responses labelled A and B, asks for a brief justification, and requires a verdict token `[[A]]`, `[[B]]`, or `[[tie]]`. The pointwise prompt presents a single response and requires a `[[score]]` token on a 1-10 scale. The prompt-sensitivity ablation (§4.7) uses a second, semantically equivalent template that labels the responses differently but requests the same verdict token.

B Data

The 29 items span 10 task categories (coding, math, ethics, reasoning, extraction, writing, knowledge, instruction, hard, and roleplay), chosen to be competitively matched so that the comparison stresses judge reliability rather than response quality. Responses A and B are single generations from GPT-4o-mini and GPT-4o respectively. For every (judge, question) pair we record each trial’s verdict, parsed score, raw completion, and resolved model snapshot, which form the basis of all metrics reported in the paper.